

A novel outlier detection approach based on formal concept analysis

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ABSTRACT

Outlier detection is a major research field for data mining. In recent years, rough set and granular computing have been successfully applied to outlier detection, and a series of excellent results have been achieved. However, as another important area of granular computing, the study of formal concept analysis applied to outlier detection has not been reported. We introduce formal concept analysis to deal with the outlier detection problem and propose a granular concept-based outlier detection model in this paper. Firstly, we define a distance measure between objects using the object belonging or not belonging to the extent of a granular concept. Secondly, a distance measure between granular concepts and an outlier degree of each granular concept are defined. And then an outlier degree of each object is constructed from the outlier degree of the granular concept by weighted average. Finally, a granular concept-based outlier detection model is given and the related algorithm is designed. Experiments are performed on fifteen public datasets, and then our algorithm is compared with classical and rough set-based outlier detection algorithms. The results of the experiments show that the proposed algorithm can effectively and efficiently detect outliers.

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1. Introduction

Outliers [1] are objects in a dataset whose characteristics are significantly different from other objects, and their appearance often implies or predicts the occurrence of events or phenomena with special significance. By analyzing the distribution characteristics of outliers, it is possible to extract abnormal information and patterns of interest from massive data. Therefore, outlier detection has become a research hotspot in the field of data mining. The purpose of outlier detection is to identify the objects different from others in large-scale data and discover potential and meaningful knowledge through data mining methods. It is now widely applied to medical processing, public safety, image processing, abnormal behavior pattern detection, trajectory anomaly detection, and other fields [2–6].

Outlier detection research first appeared in the field of statistics [7], and then Knorr et al. [8] introduced it into the field of data mining. The classical outlier detection methods are of the following four types: (1) Statistical-based method. This method [7] dates back to the 19th century and is the earliest outlier detection method. Statistical methods assume that normal objects in a dataset are produced from a probability distribution model, while

objects appearing in the low probability region are considered as outliers. (2) Distance-based detection method (DIS). It was presented by Knorr and Ng [9,10] in 1998, which defines an object as an outlier if its distance from most objects is greater than a certain distance threshold. (3) Density-based method. To address the shortcomings of previous outlier detection methods that treat outliers as a binary property (being an outlier or not), a density-based approach is put forward by Breunig et al. [11]. They considered that it makes more sense to specify a local outlier factor (LOF) for each object for many cases. The higher the LOF value of the object, the more likely it is to be an outlier. (4) Clustering-based method. Many clustering algorithms can detect outlier value, but they are mainly concerned with finding clusters. In order to find outliers, scholars have proposed some clustering-based outlier detection methods. For example, Jiang et al. [12] presented a two-stage clustering algorithm for outlier detection. He et al. [13] defined cluster-based local outliers and designed a cluster-based local outlier factor method (CBLOF) to identify these outliers. Jiang et al. [14] proposed two initialization methods for the k-Modes algorithm to select initial clusters that are not outlier initial clustering centers. Huang et al. [15] put forward a new relative outlier cluster factor (ROCF).

Granular computing is a new concept and computational paradigm in the field of artificial intelligence, which aims to granulate data into abstract concepts by simulating the cognitive mechanisms of the human brain in order to capture the potential

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knowledge that can be used by people. Since Professor Zadeh [16] proposed the concept of granular computing for the first time in 1977, many academic and applied results on granular computing have emerged. Polish scholar Pawlak [17] proposed rough set theory (RST) in 1982, which is an effective mathematical tool to process imprecise and uncertain knowledge. The essence of rough set theory is to use indistinguishable relations to form equivalence classes of objects and thus build an approximation space. Therefore, rough set theory can be seen as one of the implementations of granular computing, which highlights the importance of the notion of granular in some way.

With the development of cloud computing and big data technology, the traditional way of mining outlier information through single node computation cannot meet the growing demand of data computation. For example, DIS, kNN [18] and LOF methods are devised with the use of Euclidean distance. But, it is not reasonable to measure the similarity or dissimilarity of categorical attributes using Euclidean distance. Therefore, these methods do not work well when dealing with categorical attribute data. To make up for the shortcomings of traditional outlier detection methods, scholars have proposed outlier detection methods based on RST. Nguyen [19] proposed an approach to detect and evaluate outliers via a multi-level approximate inference scheme. Jiang et al. [20] proposed an outlier detection method with the use of rough membership function of rough sets. Shaari et al. [21] proposed a novel method for outlier detection with the use of non-reduction in RST. Xue and Liu [22] proposed a semi-supervised approach for detecting outliers based on RST. Macia-Perez et al. [23] developed the mathematical framework for detecting outliers based on RST and put forward an efficient outlier detection algorithm in a large scale data. Jiang et al. [24] further proposed an outlier detection method on the basis of information entropy in RST. In the framework of rough sets, a method for detecting outliers based on classical approximation accuracy and information entropy was introduced in [25]. To effectively handle numerical or mixed attribute data, Chen et al. [26] presented a neighborhood model as a uniform framework to understand and implement neighborhood outlier detection, and then designed a neighborhood-based outlier detection algorithm. But, their detection model involves more parameters and thus can be sensitive to parameter settings. Li et al. [27] proposed a new method using the neighborhood rough set model that can deal with real-value data while maintaining the semantics of the dataset. By using neighborhood information entropy, Yuan et al. [28] put forward an approach to detect outliers in mixed attribute data. For mixed attribute data, by using fuzzy rough granule, Yuan et al. [29] defined a granular outlier degree for describing the degree of fuzzy rough granule using fuzzy approximation accuracy, and then defined an outlier degree of the object by using the granule outlier degree and designed the corresponding fuzzy rough granule-based outlier detection algorithm. Yuan et al. [30] proposed an outlier detection method based on weighted fuzzy-rough density.

In 1982, German mathematician Wille [31] proposed formal concept analysis (FCA), also known as concept lattice theory, which is a mathematical theory for expressing and handling concepts and concept hierarchies. Each node in the concept lattice is a formal concept, and a formal concept consists of two parts: an extent and an intent. The extent is an object set covered by the concept, and the intent is a set of common attributes shared by the objects of the concept. Therefore, the formal concept can be regarded as a collection of objects with common attributes, so the process of finding all formal concepts is actually the process of clustering of objects. The current researches on FCA focus on the following three areas: (1) The expansion of concept lattice models, such as attribute-oriented concept lattice [32], object-oriented concept lattice [33], fuzzy concept lattice [34], three-way

concept lattice [35,36], etc. (2) The research on the methods for constructing concept lattice, such as batch algorithm [37], incremental algorithm [38], in-close algorithm [39], etc. (3) The rule acquisition and attribute reduction of concept lattice [40]. As a powerful mathematical tool for data analysis, FCA has been effectively applied to numerous fields such as knowledge discovery, recommender systems, software engineering, etc [41–44].

With the previous analysis, we know that the methods using Euclidean distance (such as DIS, kNN and LOF methods) do not work well when processing categorical attribute data. On the other hand, some methods (such as kNN, LOF and some RST-based methods) contain parameters and the results are sensitive to these parameters. Many scholars have applied the theory of rough set into outlier detection and achieved a series of results. However, classical RST-based outlier detection methods are only applicable for dealing with categorical attribute data. As another mathematical tool of granular computational ideas, applying FCA into outlier detection has not been studied. Based on the above discussions, we apply the theory of formal concept analysis to outlier detection. Then, a novel outlier detection approach based on formal concept analysis is proposed. The research in this paper extends outlier detection methods by using formal concept analysis. It opens up a novel direction for the application of formal concept analysis. Firstly, we calculate all granule concepts of a formal context, and define a distance measure (GCD) between objects based on granule concepts. Secondly, we propose a distance measure between granule concepts by GCD. Thirdly, the outlier degree (GOD) of granule concepts is defined by the distance measure between granule concepts. Finally, we define an outlier factor (GCOF) of each object by fusing GOD, and then a related outlier detection algorithm (GCOD) is designed.

The structure of this paper is as follows: Section 2 gives a presentation of the basic knowledge about FCA and outlier detection. Section 3 defines the outlier factor of objects and designs the outlier detection algorithm (GCOD) based on FCA. The comparative experiments and analysis are in Section 4. Section 5 presents concluding remarks and future work.

2. Preliminaries

This section briefly introduces the definitions and properties needed for this paper, and more details can be found in [9,31].

2.1. Formal concept analysis

Definition 1 ([31]). A triple $\mathcal{K} = (U, M, I)$ is regarded as a formal context, where U is a nonempty finite set of objects (or samples), M is a nonempty finite set of attributes (or features), and $I \subseteq U \times M$ is a binary relation. For any $x \in U$ and $m \in M$, $(x, m) \in I$ represents that the object x has the attribute m .

A formal context $\mathcal{K} = (U, M, I)$ is regular, if it meets two requirements as follows:

- (1) For any $x \in U$, there exists $m_1, m_2 \in M$ s.t. $(x, m_1) \in I$ and $(x, m_2) \notin I$;
- (2) For any $m \in M$, there exists $x_1, x_2 \in U$ s.t. $(x_1, m) \in I$ and $(x_2, m) \notin I$.

If not otherwise specified, the formal context of our discussion is regular.

Definition 2 ([31]). Assume $\mathcal{K} = (U, M, I)$ is a formal context. For any $X \subseteq U$ and $B \subseteq M$, a pair of operators is defined as follows:

$$X^\uparrow = \{m \in M | \forall x \in X, (x, m) \in I\}, \quad (1)$$

$$B^\downarrow = \{x \in U | \forall m \in B, (x, m) \in I\}. \quad (2)$$

Table 1
A formal context $\mathcal{K}$.

U	m_1	m_2	m_3	m_4
x_1	$\times$	$\times$		$\times$
x_2			$\times$	
x_3	$\times$	$\times$		$\times$
x_4	$\times$		$\times$	$\times$
x_5	$\times$	$\times$		

Table 2
The value sets under f and g .

U	f	g
x_1	0.1	0.5
x_2	0.2	0.6
x_3	0.3	0.5
x_4	0.2	0.7
x_5	0.2	0.5
x_6	0.7	0.1

Where $X^\uparrow$ is an attribute set held by all objects of X , and $B^\downarrow$ consists of the objects which possess every attribute of B . For $X, X_1, X_2 \subseteq U$ and $B, B_1, B_2 \subseteq M$, the pair of operators $\uparrow, \downarrow$ possesses some properties as follows [31]:

- (1) $X_1 \subseteq X_2 \Rightarrow X_2^\uparrow \subseteq X_1^\uparrow, B_1 \subseteq B_2 \Rightarrow B_2^\downarrow \subseteq B_1^\downarrow$;
- (2) $X \subseteq X^{\uparrow\downarrow}, B \subseteq B^{\downarrow\uparrow}$;
- (3) $X^\uparrow = X^{\uparrow\downarrow\uparrow}, B^\downarrow = B^{\downarrow\uparrow\downarrow}$;
- (4) $X \subseteq B^\downarrow \Leftrightarrow B \subseteq X^\uparrow$;
- (5) $(X_1 \cup X_2)^\uparrow = X_1^\uparrow \cap X_2^\uparrow, (B_1 \cup B_2)^\downarrow = B_1^\downarrow \cap B_2^\downarrow$.

Definition 3 ([31]). Assume $\mathcal{K} = (U, M, I)$ is a formal context. For $X \subseteq U$ and $B \subseteq M$, we call a pair (X, B) formal concept (or concept), if $X^\uparrow = B$ and $B^\downarrow = X$, where X and B are referred to as the extent and intent of concept (X, B) , respectively.

All concepts of formal context $\mathcal{K}$ form a complete lattice denoted by $L(\mathcal{K})$, where for any $(X, B), (Y, C) \in L(\mathcal{K})$, the order relation $\leq$ is defined as:

$$(X, B) \leq (Y, C) \Leftrightarrow X \subseteq Y \Leftrightarrow C \subseteq B. \quad (3)$$

In addition, the infimum and supremum of $L(\mathcal{K})$ are given by:

$$(X, B) \wedge (Y, C) = (X \cap Y, (B \cup C)^{\downarrow\uparrow}), \quad (4)$$

$$(X, B) \vee (Y, C) = ((X \cup Y)^{\uparrow\downarrow}, B \cap C). \quad (5)$$

Example 1. Let us discuss a formal context $\mathcal{K} = (U, M, I)$ displayed by Table 1, where $U = \{x_1, x_2, \dots, x_5\}$, $M = \{m_1, m_2, m_3, m_4\}$, $\times$ represents the object has the attribute. By computing, we can get

$$L(\mathcal{K}) = \{(\emptyset, M), (x_4, m_1 m_3 m_4), (x_1 x_3, m_1 m_2 m_4), (x_2 x_4, m_3), (x_1 x_3 x_5, m_1 m_2), (x_1 x_3 x_4, m_1 m_4), (x_1 x_3 x_4 x_5, m_1), (U, \emptyset)\}.$$

2.2. Outlier detection

On the basis of definitions given by Hawkin [1], the outlier is formally defined in the following.

Definition 4 ([29]). For a given set of objects $U = \{x_1, x_2, \dots, x_n\}$. Let $F = \{f(x_1), f(x_2), \dots, f(x_n)\}$ and $G = \{g(x_1), g(x_2), \dots, g(x_n)\}$ are two value sets of U , produced from two mechanisms $f(\cdot)$ and $g(\cdot)$, respectively. Denote $M(\cdot)$ as a measure that represents characterization of an object. For $x_i \in U$ and $f(x_i) \neq g(x_i)$, we call x_i is an outlier in U , if for any $x_j \in U - \{x_i\}$, it holds that $M(x_i) \gg M(x_j)$, where $M(\cdot)$ is associated with the mechanisms $f(\cdot)$ and $g(\cdot)$.

Then Knorr and Ng [9] put forward a specific definition of distance-based outliers.

Definition 5 ([9]). For a given set of objects $U = \{x_1, x_2, \dots, x_n\}$. Let δ ($\delta \geq 0$) and η ($\eta \geq 0$) be a given distance threshold and fraction threshold, respectively. For any $x \in U$, a function $\theta : U \rightarrow [0, 1]$ is determined by:

$$\theta(x) = \frac{|\{x' \in U | \text{dist}(x, x') \leq \delta\}|}{|U|}, \quad (6)$$

Table 3
The value of the object under θ .

U	θ
x_1	0.6667
x_2	0.8333
x_3	0.6667
x_4	0.5000
x_5	0.8333
x_6	0.1667

where $|\cdot|$ is the cardinality of a classical set and $\text{dist}(\cdot, \cdot)$ is a given distance metric. Then x is regarded as an outlier, if $\theta(x) \leq \eta$.

Example 2. Let $U = \{x_1, x_2, \dots, x_6\}$ be a set of objects, while f, g are two mechanisms given by Table 2. For any $x, y \in U$, a distance metric is defined as $\text{dist}(x, y) = \sqrt{(f(x) - f(y))^2 + (g(x) - g(y))^2}$. Let $\delta = 0.2$. By Eq. (6), for any $x \in U$, we can get $\theta(x)$ shown by Table 3. Therefore, when $\eta = 0.2$, x_6 is an outlier. From Table 2, we have $f(x_6) = 0.7$, while other objects take values less than 0.3 under f , and $g(x_6) = 0.1$, while other objects take values greater than 0.5 under g . Therefore, it is reasonable that x_6 is an outlier.

3. A novel method to detect outliers based on FCA

In recent years, scholars have investigated outlier detection methods based on granular computing and RST to compensate for the shortcomings of traditional outlier detection methods. Although these methods have demonstrated their effectiveness for detecting outliers. However, it is noteworthy that some methods need to obtain equivalence classes based on equivalence relations, and the uniqueness of equivalence classes and approximation sets leads to some information in the original dataset being ignored, which leads to unsatisfactory detection results. As another important mathematical tool for data mining, FCA has been applied successfully in information retrieval, knowledge discovery, association analysis, software engineering and other fields. Next, we will present an outlier detection model on the basis of FCA and further design a related outlier detection algorithm.

3.1. An outlier detection model based on granular concepts

Definition 6 ([45]). Let $\mathcal{K} = (U, M, I)$ be a formal context. For any $x \in U$ and $m \in M$, $(\{x\}^{\uparrow\downarrow}, \{x\}^{\uparrow})$ and $(\{m\}^\downarrow, \{m\}^{\uparrow\downarrow})$ (or $(x^{\uparrow\downarrow}, x^\uparrow)$ and $(m^\downarrow, m^{\uparrow\downarrow})$ for short) are both formal concepts, called object granular concept and attribute granular concept, respectively. The set of all object (attribute) granular concepts generated by $\mathcal{K}$ denotes as $L_{og}(\mathcal{K})$ ($L_{ag}(\mathcal{K})$), the set of corresponding extents records as $Ext_{Log}(\mathcal{K})$ ($Ext_{Lag}(\mathcal{K})$).

Theorem 1 ([45]). Let $\mathcal{K} = (U, M, I)$ be a formal context. For any $(X, B) \in L(\mathcal{K})$, it holds that

$$(X, B) = \bigvee_{x \in X} (x^{\uparrow\downarrow}, x^\uparrow) = \bigwedge_{m \in B} (m^\downarrow, m^{\uparrow\downarrow}).$$

From [Theorem 1](#), we can get that for a given formal context $\mathcal{K}$, any concept of $\mathcal{K}$ can be generated by some object granular concepts (or attribute granular concepts).

For any $x \in U$, the traditional outlier detection methods only determine a binary property on x , i.e., whether x is an outlier or not. However, for practical problems, it is more reasonable to specify an outlier factor to x . Based on this perspective, we put forward the following definitions to determine the outlier factor for each object based on FCA.

Definition 7. Assume $\mathcal{K} = (U, M, I)$ is a formal context. For any $x, y \in U$, a granular concept-based distinguishing degree (GCD) between x and y is given by:

$$GCD(x, y) = |\{(X, B) \in L | (x \in X \wedge y \notin X) \vee (x \notin X \wedge y \in X)\}|, \quad (7)$$

where $L = L_{og}(\mathcal{K}) \cup L_{ag}(\mathcal{K})$.

Note that in the Formula (7), we use the number of granular concepts that can distinguish between two objects (that is, the number of granular concepts whose extents contain one object and not the other) to determine the degree of distinction of two objects. But why use granular concepts instead of formal concepts? There are two reasons for this. (1) Constructing the concept lattice $L(\mathcal{K})$ of a formal context $\mathcal{K}$ is NP-hard, while the complexity of computing all granular concepts is $O(|U| + |M|)$, and it is obviously faster to obtain all granular concepts, which can improve the applicability of the model. (2) From [Theorem 1](#), we can know that any concept can be generated by some granular concepts, the granular concept can be seen as the basic granular for constructing the concept lattice.

Theorem 2. The distinguishing degree GCD defined by [Definition 7](#) is a distance measure.

Proof. We need to verify that GCD satisfies the three axioms of distance. Assume $\mathcal{K} = (U, M, I)$ is a formal context.

(1) Non-negative. For any $x, y \in U$, we can obviously obtain $GCD(x, y) \geq 0$.

(2) Symmetry. For any $x, y \in U$, it holds that $GCD(x, y) = |\{(X, B) \in L | (x \in X \wedge y \notin X) \vee (x \notin X \wedge y \in X)\}| = |\{(X, B) \in L | (y \in X \wedge x \notin X) \vee (y \notin X \wedge x \in X)\}| = GCD(y, x)$.

(3) Direct transmission. For any $x, y, z \in U$, we can get

$$GCD(x, y) = |\{(X, B) \in L | (x \in X \wedge y \notin X) \vee (x \notin X \wedge y \in X)\}|$$

and

$$\begin{aligned} GCD(x, z) + GCD(y, z) \\ = |\{(X, B) \in L | (x \in X \wedge z \notin X) \vee (x \notin X \wedge z \in X)\}| \\ + |\{(X, B) \in L | (y \in X \wedge z \notin X) \vee (y \notin X \wedge z \in X)\}| \end{aligned}$$

by Formula (7). Obviously, it can be obtained $((x \in X \wedge y \notin X) \vee (x \notin X \wedge y \in X)) \rightarrow (x \in X \vee y \in X)$ and $((x \in X \wedge y \notin X) \vee (x \notin X \wedge y \in X)) \rightarrow (x \notin X \vee y \notin X)$, then we get

$$\begin{aligned} GCD(x, z) + GCD(y, z) \\ = |\{(X, B) \in L | (x \in X \wedge z \notin X) \vee (x \notin X \wedge z \in X)\}| \\ + |\{(X, B) \in L | (y \in X \wedge z \notin X) \vee (y \notin X \wedge z \in X)\}| \\ = |\{(X, B) \in L | (x \in X \wedge z \notin X) \vee (x \notin X \wedge z \in X) \\ \vee (y \in X \wedge z \notin X) \vee (y \notin X \wedge z \in X)\}| \\ = |\{(X, B) \in L | (z \notin X \wedge (x \in X \vee y \in X)) \\ \vee (z \in X \wedge (x \notin X \vee y \notin X))\}| \\ \geq |\{(X, B) \in L | (z \notin X \wedge ((x \in X \wedge y \notin X) \vee (x \notin X \wedge y \in X))) \\ \vee (z \in X \wedge ((x \in X \wedge y \notin X) \vee (x \notin X \wedge y \in X)))\}| \\ = |\{(X, B) \in L | (z \notin X \vee z \in X)\}| \end{aligned}$$

$$\begin{aligned} & \wedge ((x \in X \wedge y \notin X) \vee (x \notin X \wedge y \in X))\} \\ & = |\{(X, B) \in L | (x \in X \wedge y \notin X) \vee (x \notin X \wedge y \in X)\}| \\ & = GCD(x, y) \end{aligned}$$

In summary, the distinguishing degree GCD defined by [Definition 7](#) is a distance measure. $\square$

In the following we define a distinguishing degree between the granular concepts by the GCD between objects.

Definition 8. Let $\mathcal{K} = (U, M, I)$ be a formal context, and $L = L_{og}(\mathcal{K}) \cup L_{ag}(\mathcal{K})$. For any $(X_1, B_1), (X_2, B_2) \in L$, the distinguishing degree between two granular concepts is defined by:

$$\begin{aligned} M((X_1, B_1), (X_2, B_2)) \\ = \begin{cases} \frac{\sum_{x \in X_1, y \in X_2} GCD(x, y)}{|X_1| \cdot |X_2|}, & \text{if } (X_1, B_1) \neq (X_2, B_2) \\ 0, & \text{if } (X_1, B_1) = (X_2, B_2) \end{cases} \quad (8) \end{aligned}$$

Theorem 3. The distinguishing degree M between two granular concepts defined by [Definition 8](#) is a distance measure.

Proof. We need to verify that M satisfies the three axioms of distance. Let $\mathcal{K} = (U, M, I)$ be a formal context, and $L = L_{og}(\mathcal{K}) \cup L_{ag}(\mathcal{K})$.

(1) Non-negative. For any $(X_1, B_1), (X_2, B_2) \in L$, it is obvious that $M((X_1, B_1), (X_2, B_2)) \geq 0$.

(2) Symmetry. For any $(X_1, B_1), (X_2, B_2) \in L$, if $(X_1, B_1) = (X_2, B_2)$, it is obvious that $M((X_1, B_1), (X_2, B_2)) = 0 = M((X_2, B_2), (X_1, B_1))$. On other hand, if $(X_1, B_1) \neq (X_2, B_2)$, it can be obtained

$$\begin{aligned} M((X_1, B_1), (X_2, B_2)) &= \frac{\sum_{x \in X_1, y \in X_2} GCD(x, y)}{|X_1| \cdot |X_2|} \\ &= \frac{\sum_{y \in X_2, x \in X_1} GCD(y, x)}{|X_2| \cdot |X_1|} = M((X_2, B_2), (X_1, B_1)) \end{aligned}$$

by the symmetry of the distance measure GCD.

(3) Direct transmission. For any $(X_1, B_1), (X_2, B_2), (X_3, B_3) \in L$, if $(X_3, B_3) = (X_1, B_1)$ or $(X_3, B_3) = (X_2, B_2)$, then we can get $M((X_1, B_1), (X_3, B_3)) = 0$ or $M((X_2, B_2), (X_3, B_3)) = 0$, it follows that

$$M((X_1, B_1), (X_2, B_2)) = M((X_1, B_1), (X_3, B_3)) + M((X_2, B_2), (X_3, B_3)).$$

On the other hand, if $(X_1, B_1) \neq (X_3, B_3)$ and $(X_2, B_2) \neq (X_3, B_3)$, we have

$$\begin{aligned} & |X_1| \cdot |X_2| \cdot |X_3| M((X_1, B_1), (X_3, B_3)) \\ & + |X_1| \cdot |X_2| \cdot |X_3| M((X_2, B_2), (X_3, B_3)) \\ & = |X_2| \sum_{x \in X_1, z \in X_3} GCD(x, z) + |X_1| \sum_{y \in X_2, z \in X_3} GCD(y, z) \\ & = |X_2| \sum_{x \in X_1} \sum_{z \in X_3} GCD(x, z) + |X_1| \sum_{y \in X_2} \sum_{z \in X_3} GCD(y, z) \\ & = \sum_{z \in X_3} (|X_2| \sum_{x \in X_1} GCD(x, z) + |X_1| \sum_{y \in X_2} GCD(y, z)) \\ & \geq \sum_{z \in X_3} \sum_{x \in X_1, y \in X_2} GCD(x, y) \\ & \quad \text{(by the direct transmission of the distance measure GCD)} \\ & = |X_3| \sum_{x \in X_1, y \in X_2} GCD(x, y) \\ & = |X_1| \cdot |X_2| \cdot |X_3| M((X_1, B_1), (X_2, B_2)) \end{aligned}$$

It follows that $M((X_1, B_1), (X_3, B_3)) + M((X_2, B_2), (X_3, B_3)) \geq M((X_1, B_1), (X_2, B_2))$.

In summary, the distinguishing degree M defined by Definition 8 is a distance measure. $\square$

Theorem 3 indicates that the distinguishing degree between two granular concepts given by Formula (8) is a distance measure. Therefore it is reasonable to use Formula (8) to denote the degree of distinguishing between granular concepts.

Definition 9. Let $\mathcal{K} = (U, M, I)$ be a formal context, and $L = L_{og}(\mathcal{K}) \cup L_{ag}(\mathcal{K})$. For any $(X, B) \in L$, the outlier degree (GOD) of granular concept (X, B) is given by:

$$GOD(X, B) = \frac{\sum_{i=1}^{|L|} M((X, B), (X_i, B_i))}{|L|}, \quad (9)$$

The outlier degree of a granular concept is the sum of the distance measure between the granular concept and others over the number of granular concepts, that is, when the distance measure between a granular concept and all other granular concepts is larger, the outlier degree of the granular concept is also larger.

Definition 10. Assume $\mathcal{K} = (U, M, I)$ is a formal context, and $L = L_{og}(\mathcal{K}) \cup L_{ag}(\mathcal{K})$. For any $x \in U$, the granular concept-based outlier factor (GCOF) of x is defined by:

$$GCOF(x) = \frac{\sum_{j=1}^{m_x} (GOD(X_j, B_j) \times W(X_j))}{m_x}, \quad (10)$$

where $m_x = |\{(X, B) \in L | x \in X\}|$, and $W(X_j) = 1 - \sqrt[3]{\frac{|X_j|}{|U|}}$.

With respect to the weight function $W(X_j)$, since outlier detection usually concerns relatively few objects in a data set, the smaller the cardinality of the extents of the granular concepts is, the larger the weight is. We rank the objects according to GCOF from largest to smallest, and the higher ranked objects are more probably to become outliers.

So far, an outlier detection model (GCOD) based on FCA have been developed. The line of constructing the model consists of five steps: (1) Discretizing the data and then transforming it into a formal context $\mathcal{K}$. (2) Computing the granular concept set $L = L_{og}(\mathcal{K}) \cup L_{ag}(\mathcal{K})$. (3) Computing the distance measure GCD of any two objects. (4) Computing the distance measure of any two granular concepts, and then obtaining the GOD of granular concepts. (5) The outlier factor GCOF of each object can be obtained, and then perform outlier detection.

Example 3. We continue our discussion with Example 2. Regarding the two mechanisms f and g as two attributes, we need to first discretize the data since the object takes values on the $[0, 1]$ interval under the attribute. We use equal width discretization method(EW), the attribute values are transformed into discrete attributes with discretization step to 4, so that the Table 2 can be transformed into the Table 4, and then translates Table 4 into a regular formal context $\mathcal{K}$ shown in Table 5. After calculation, we have

$$L = L_{og}(\mathcal{K}) \cup L_{ag}(\mathcal{K}) = \{(x_3, f_2g_3), (x_6, f_4g_1), (x_1x_5, f_1g_3), (x_2x_4, f_1g_4), (x_1x_3x_5, g_3), (x_1x_2x_4x_5, f_1)\}.$$

The GCD between objects is shown by Table 6. Then, the GOD of each granular concept can be calculated as shown in Table 7. Finally, we can get the outlier factor GCOF of each object using Eq. (10), it is shown in Table 8. The larger the outlier factor GCOF of an object, the more probably it can be an outlier. Therefore x_6 is most likely to be an outlier, which is in agreement with the conclusion in Example 2.

Table 4

Data after discretization.

U	f	g
x_1	1	3
x_2	1	4
x_3	2	3
x_4	1	4
x_5	1	3
x_6	4	1

Table 5

Formal context $\mathcal{K}$.

U	f_1	f_2	f_4	g_1	g_3	g_4
x_1	×				×	
x_2	×					×
x_3		×			×	
x_4	×					×
x_5	×				×	
x_6			×	×		

Table 6

GCD.

	x_1	x_2	x_3	x_4	x_5	x_6
x_1	0	3	3	3	0	4
x_2	3	0	4	0	3	3
x_3	3	4	0	4	3	3
x_4	3	0	4	0	3	3
x_5	0	3	3	3	0	4
x_6	4	3	3	3	4	0

Table 7

GOD.

	GOD
(x_3, f_2g_3)	2.5833
(x_6, f_4g_1)	2.8611
(x_1x_5, f_1g_3)	2.0833
(x_2x_4, f_1g_4)	2.4722
$(x_1x_3x_5, g_3)$	2.2500
$(x_1x_2x_4x_5, f_1)$	2.2778

Table 8

GCOF of each object.

	x_6	x_3	x_2	x_4	x_1	x_5
GCOF	1.2866	0.8129	0.5230	0.5230	0.4637	0.4637

3.2. Specific algorithm

The previous section gives the outlier factor GCOF of each object based on FCA, this section puts forward Algorithm 1 (GCOD) for outlier detection. In this algorithm, the input is “an original data set” and the output is “the outlier factor GCOF of each object”. The outlier factor GCOF of each object is obtained by six “for” loops. First two “for” loops (steps 3–8) calculates L (the granular concept set of the formal context $\mathcal{K} = (U, M, I)$), the frequency of it is $|U| + |M|$. The third “for” loop (steps 10–12) calculates GCD between any two objects, and the number of loops is $|U|^2$. The fourth “for” loop (steps 13–15) calculates the distance measure between any two granular concepts, and the number of loops is $|L|^2$. The fifth “for” loop (steps 16–18) calculates GOD of each granular concept, and the number of loops is $|L|$. The sixth “for” loop (steps 19–21) calculates GCOF of each object, and the number of loops is $|U| \times |L|$. In summary, in the worst-case scenario, the time complexity of Algorithm 1 is $O(|U|^2 + |M| + |L|^2 + |U| \times |L|)$. Moreover, from $L = L_{og}(\mathcal{K}) \cup L_{ag}(\mathcal{K}) = \{(x^{\uparrow\downarrow}, x^{\uparrow}), (m^{\downarrow}, m^{\uparrow\downarrow}) | x \in U, m \in M\}$, we have $|L| \leq |L_{og}(\mathcal{K})| + |L_{ag}(\mathcal{K})| \leq |U| + |M|$. Therefore, the time complexity of Algorithm 1 is $O(|U|^2 + |M|^2 + |U| \times |M|)$. For Algorithm FIED

Table 9
Basic information on datasets.

NO.	Data set	Abbreviation	Attribute		Outlier	Normal
			Numerical	Categorical		
1	Diabetes	Diab	8	0	26	500
2	Pima	Pima	9	0	55	500
3	Sonar, mines vs. rocks	Sonar	60	0	10	97
4	Wisconsin Diagnostic breast cancer	Wdbc	31	0	39	357
5	Wine	Wine	13	0	10	119
6	Lymphography	Lymp	0	18	6	142
7	Vote_republican_29_variant1	Vote	0	16	29	296
8	Audiology_variant1	Audiology	0	69	53	226
9	Monks_0_25_variant1	Monks	0	6	25	253
10	Chess_nowin_16_variant1	Chess	0	36	16	1685
11	Hepatitis_2_9_variant1	Hepatitis	6	13	9	94
12	Credit approval	Cred	6	9	42	383
13	Heart disease	Heart	6	7	16	150
14	Horse	Horse	8	19	12	244
15	German	Germ	7	13	36	700

(fuzzy information entropy-based [46]), the time complexity is $O(|U|^2 \times |M|)$. Obviously, the time complexity of our proposed algorithm is lower.

Remark 1. In Step 1, we get a formal context from an original data set. Specifically, for numerical attributes, they need to be discretized. Then, for each attribute, each attribute value is regarded as a new attribute, object with this new attribute value is marked as 1. By this way, we can obtain a formal context.

Algorithm 1 GCOD Algorithm

Input: An original data set

Output: The outlier factor GCOF of each object

```

1: Data preprocessing: transforming the original data set into a
   regular formal context  $\mathcal{K} = (U, M, I)$ ;
2: Initialize  $L = \emptyset$ ,  $L_{og}(\mathcal{K}) = \emptyset$  and  $L_{ag}(\mathcal{K}) = \emptyset$ ;
3: for  $x \in U$  do
4:    $L_{og} = L_{og} \cup \{(x^{\downarrow}, x^{\uparrow})\}$ ;
5: end for
6: for  $m \in M$  do
7:    $L_{ag} = L_{ag} \cup \{(m^{\downarrow}, m^{\uparrow})\}$ ;
8: end for
9:  $L = L_{og} \cup L_{ag}$ ;
10: for  $x, y \in U$  do
11:   Calculate  $GCD(x, y)$  by Formula (7);
12: end for
13: for  $(X_1, B_1), (X_2, B_2) \in L$  do
14:   Calculate  $M((X_1, B_1), (X_2, B_2))$  by Formula (8);
15: end for
16: for  $(X, B) \in L$  do
17:   Calculate  $GOD(X, B)$  by Formula (9);
18: end for
19: for  $x \in U$  do
20:   Calculate  $GCOF(x)$  by Formula (10);
21: end for

```

4. Experiments and comparative analysis

To assess the efficiency of the proposed algorithm GCOD, we compare its performance with algorithms distance-based (DIS) [10], local outlier factor (LOF) [11], information entropy-based (IE) [24], corresponding fuzzy rough granules-based (FR-GOD) [29], neighborhood information entropy-based outlier detection (NIEOD) [28], and fuzzy information entropy-based (FIEOD) [46]. Due to the fact that algorithms DIS and LOF are designed based on numerical attribute data, categorical attribute data need to be standardized for these algorithms. Meanwhile,

numerical attribute datasets need to be pre-processed by discretization for algorithms IE and GCOD.

4.1. Experimental datasets and pre-processing

We select fifteen datasets from publicly available web pages,^{1,2} which are frequently used in several outlier detection research works to evaluate the detection performance. To perform outlier detection, few datasets in the UCI exist directly. These available datasets are generated from some datasets in the UCI. The specific generating process can be found in web pages.^{1,2} The basic information of these datasets are shown in Table 9. It can be seen that five datasets contain only numerical attributes, five dataset contains only categorical attributes, and the remaining five datasets consist of numerical and categorical attributes, which are mixed attributes. In addition, the datasets have objects between 103 and 1701, while the attributes range from 8 to 69.

4.2. Experimental setup

Some algorithms contain parameters and the results are sensitive to these parameters based on different datasets. Therefore, before starting our comparison experiments, we first select the optimal parameters for these algorithms. Specifically, the algorithm LOF have parameters $MinPts$. Therefore in our experiments we repeat the algorithms in the range of $[1, |U|/4]$ with steps of 1 and select the optimal parameters for our comparison algorithms. For algorithms IE and the proposed GCOD, datasets need to be discretized, and here we choose the discretization method based on Equal Width (EW). For the algorithms FIEOD, FRGOD and NIEOD, the experiment is repeated on the fifteen datasets and the optimal parameters λ , θ and σ in the range of $[0.1, 5]$ with steps of 0.1 are selected as the comparison algorithms. Eventually, the optimal parameter selection and discretization step size based on different datasets are shown in Table 10. Moreover, in traditional DIS methods, it is unreasonable that outliers are treated as binary classification, and there are two parameters, the selection of parameters cannot be determined. Therefore, this paper uses strategy in documents [47] do define the distance-based outlier factor $DOF(x) = \sum_{y \in U} dist(x, y)$ for each object.

To estimate the performance of different outlier detection methods, this paper focuses on precision (P), recall (R), receiver operating characteristic (ROC) curves and AUC values. They are described in the following.

¹ <https://github.com/BELLoney/Outlier-detection>

² <http://odds.cs.stonybrook.edu>

Table 10
Optimal parameter selection and discretization step.

Data set	MinPts(LOF)	λ (FIEOD)	θ (FRGOD)	σ (NIEOD)	Discretization step
Diab	7	2.8	2.9	1.6	4
Pima	136	0.9	2.5	1.2	6
Sonar	13	1	1.2	0.6	2
Wdbc	74	0.6	0.9	1.4	4
Wine	27	2.7	2.8	2.7	2
Lymph	36	–	–	–	–
Vote	72	–	–	–	–
Audiology	32	–	–	–	–
Monks	39	–	–	–	–
Chess	3	–	–	–	–
Hepatitis	28	2.3	1.4	0.9	3
Cred	5	1	1.3	0.9	5
Heart	41	3.9	1.9	1.9	6
Horse	15	1.4	1.7	2.8	6
Germ	177	0.6	0.7	2.8	9

In these algorithms, the final output is the outlier factor of each object. Therefore, the higher the outlier factor is, the more probably it is to be an outlier. The objects are sorted in descending order based on the outlier factor and numbered from 1. For a specific t , the objects with serial numbers less than or equal to t are selected as the outliers obtained from the experiment. Obviously, if t is chosen to be small, some of the real outliers will be ignored. Conversely, if a larger t is selected, more objects will be judged as outliers, which leads to too many misclassifications. Therefore, we change the value of t and use precision (P) and recall (R) to measure the performance of the algorithms. For any $t \in \{1, 2, \dots, |U|\}$, let $OS(t)$ denote the collection of outliers chosen by a specific t , that is to say $OS(t)$ is the set of top t objects, after sorted in descending order based on the outlier factor. Obviously $OS(t)$ varies with t . The OS° denotes the set of real outliers in the dataset, and the precision $P(t)$ and recall $R(t)$ are defined as follows:

$$P(t) = \frac{|OS(t) \cap OS^\circ|}{t} \cdot 100\%, \quad (11)$$

$$R(t) = \frac{|OS(t) \cap OS^\circ|}{|OS^\circ|} \cdot 100\%, \quad (12)$$

where precision $P(t)$ denotes proportion of real outliers to outliers by experimentally detected at a specific t ; and recall $R(t)$ denotes the proportion of real outliers by experimentally detected at a specific t to all real outliers.

Meanwhile the false positive rate (FPR) and the true positive rate (TPR) are defined in the following:

$$FPR(t) = \frac{|OS(t) - OS^\circ|}{|U - OS^\circ|} \cdot 100\%, \quad (13)$$

$$TPR(t) = R(t) = \frac{|OS(t) \cap OS^\circ|}{|OS^\circ|} \cdot 100\%, \quad (14)$$

where the false positive rate $FPR(t)$ represents the proportion of non-real outliers by experimentally detected at a given t to the total number of non-real outliers.

The AUC value is given by

$$AUC = \frac{1}{|OS^\circ| \parallel U - OS^\circ|} \sum_{x_i \in OS^\circ, x_j \in (U - OS^\circ)} P_{ij}, \quad (15)$$

Where if $S(x_i) > S(x_j)$, then $P_{ij} = 1$; if $S(x_i) = S(x_j)$, then $P_{ij} = \frac{1}{2}$; if $S(x_i) < S(x_j)$, then $P_{ij} = 0$. In addition, for any $x \in U$, $S(x)$ means the outlier factor of x .

Remark 2. When the value of t is given, the larger the values of $P(t)$, $R(t)$ and $TPR(t)$ are, the more effective the outlier detection effect is. Meanwhile, the smaller the value of $FPR(t)$ is, the more

effective the outlier detection effect is. In particular, the values of $P(t)$ and $R(t)$ are equal when $t = |OS^\circ|$. The ROC curve is a curve plotted with the false positive rate $FPR(t)$ as the horizontal coordinate and the true positive rate $TPR(t)$ as the vertical coordinate. The ROC curve is employed to judge the performance of different algorithms for detecting outliers, and the performance is better when the curve is closer to the upper left corner of the first quadrant, i.e., the larger the area under the curve. The value range of AUC is $[0, 1]$. The larger the AUC value of an outlier detection algorithm, the better its performance.

4.3. Experimental results and analysis

The running times and results of the comparison experiments are shown in Tables 11–15, and the ROC curves are shown in Figs. 1–3.

As can be seen from Table 12, the proposed algorithm GCOD exhibits superior performance in handling outlier detection for numerical attribute datasets. For example, for the dataset Wine and $t = 10$, the precision $P(t)$ of the algorithm GCOD is 50%, while based on the algorithms DIS, LOF, IE, FIEOD, FRGOD and NIEOD, their $P(t)$ are 20%, 40%, 30%, 20%, 20% and 20%, respectively. It is obvious that the value of $P(t)$ of the algorithm GCOD is greater than that of other algorithms, when $t = 10$. On other hand, when the recall $R(t)$ equals to 100% for the first time, the algorithm GCOD's t is 24, while based on the algorithms DIS, LOF, IE, FIEOD, FRGOD and NIEOD, their t are 88, 24, 29, 47, 54 and 49, respectively. It indicates that the algorithm GCOD can detect all 10 real outliers when $t = 24$, and the algorithms except the algorithm LOF require a larger t than the algorithm GCOD to detect all the outliers in this dataset. It should be noted that for the dataset Wdbc, the algorithm GCOD and the algorithm FIEOD have the same results and both outperform the other algorithms, but as seen from the running time of Table 11, the algorithm FIEOD takes 87.4531 s while the algorithm GCOD only takes 3.2188 s, which means that the running time of the algorithm GCOD is much lower than the algorithm FIEOD, so the algorithm GCOD is better than the algorithm FIEOD under this data set. Moreover, the average value of $P(t)$ and $R(t)$ under the algorithm GCOD is greater than the other algorithms.

Based on categorical attribute datasets, the experimental results are recorded in Table 13. For the datasets Lymph, Audiology and Chess, the average precision $P(t)$ and recall $R(t)$ of the algorithm GCOD are larger than that of the other algorithms. While the recall $R(t)$ achieves 100% for the first time, t for GCOD is smaller than that of other algorithms under datasets Lymph, Vote and Chess. Meanwhile, for datasets Monks and Audiology, t for GCOD is smaller than that of all other algorithms except algorithms FIEOD, FRGOD and NIEOD, respectively. But the running time of the proposed algorithm GCOD is much shorter than that of algorithms FIEOD, FRGOD and NIEOD under datasets Monks and Audiology, respectively.

For mixed attribute datasets, the experimental results are recorded in Table 14. Our proposed algorithm GCOD works better with most datasets, or only second to algorithms FIEOD, FRGOD and NIEOD. But these algorithms have parameters, which requires optimal parameter selection first, and is much longer than algorithm GCOD in terms of running time.

The effectiveness and efficiency of the algorithm GCOD can also be visualized from Figs. 1–3 and Table 15. By Figs. 1–3, we have that the ROC curves of the algorithm GCOD are closet to the upper left corner of the graph in eight datasets and closer to the upper left corner of the graph in remaining datasets. From Table 15, some corresponding analyses are as follows. For 15 datasets, algorithm GCOD has the highest AUC values in 8 datasets, while algorithms LOF, IE, FIEOD and NIEOD obtain the

Table 11
Running times (s).

	DIS	LOF	IE	FIEOD	FRGOD	NIEOD	GCOD
Diab	0.1094	0.2500	14.2950	53.6718	26.9986	27.8336	1.6094
Pima	0.1875	0.2188	21.8476	62.3906	35.1804	41.7532	3.2031
Sonar	0.2926	0.1875	1.8037	3.2656	1374	2.7565	0.8281
Wdbc	0.1250	0.2188	24.1707	87.4531	1722.5	44.4785	3.2188
Wine	0.0938	0.1406	3.5469	0.9219	12.5703	0.5441	0.3750
Lymph	0.2445	0.2429	1.4467	1.2169	1.6233	1.1384	0.4882
Vote	0.4300	0.4091	4.7450	5.4617	1.2755	10.6610	0.6896
Audiology	0.4078	0.4138	9.8758	11.5376	63.1564	22.9542	2.0710
Monks	0.4334	0.4101	1.7157	1.6773	0.2694	1.3275	0.5380
Chess	0.5976	0.7799	2478	4700	271.1777	9041.8	172.2458
Hepatitis	0.4214	0.4202	0.8802	0.8327	6.6656	0.4632	0.3125
Cred	0.2831	0.2933	13.1425	22.6038	43.6713	34.8275	1.8750
Heart	0.2949	0.2973	1.2598	1.3281	4.5443	1.1580	0.4375
Horse	0.2901	0.2896	6.3103	6.0221	44.6650	11.2145	1.5531
Germ	0.0625	0.3151	101.1626	121.2092	123.7666	194.6356	9.1875

Table 12
Comparison of experimental results for numerical attribute datasets (%).

Data Set	t	DIS		LOF		IE		FIEOD		FRGOD		NIEOD		GCOD	
		P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)
Diab	16	56.25	34.62	50.00	30.77	75.00	46.15	68.75	42.31	75.00	46.15	81.25	50.00	68.75	42.31
	26	42.31	42.31	30.77	30.77	69.23	69.23	69.23	69.23	69.23	69.23	69.23	69.23	73.08	73.08
	39	33.33	50.00	41.03	61.54	51.28	76.92	56.41	84.62	64.10	96.15	58.97	88.46	56.41	84.62
	49	32.65	61.54	40.82	76.92	44.90	84.62	48.98	92.31	53.06	100.00	51.02	96.15	46.94	88.46
	52	34.62	69.23	40.38	80.77	42.31	84.62	46.15	92.31	50.00	100.00	50.00	100.00	44.23	88.46
	58	34.48	76.92	36.21	80.77	37.93	84.62	44.83	100.00	44.83	100.00	44.83	100.00	44.83	100.00
	91	28.57	100.00	26.37	92.31	26.37	92.31	28.57	100.00	28.57	100.00	28.57	100.00	28.57	100.00
	98	26.53	100.00	26.53	100.00	24.49	92.31	26.53	100.00	26.53	100.00	26.53	100.00	26.53	100.00
	160	16.25	100.00	16.25	100.00	16.25	100.00	16.25	100.00	16.25	100.00	16.25	100.00	16.25	100.00
	Average	33.89	70.51	34.26	72.65	43.08	81.20	45.08	86.75	47.51	90.17	47.41	89.32	45.07	86.32
Pima	20	55.00	20.00	65.00	23.64	70.00	25.45	75.00	27.27	75.00	27.27	75.00	27.27	75.00	27.27
	40	52.50	38.18	47.50	34.55	67.50	49.09	67.50	49.09	70.00	50.91	67.50	49.09	67.50	49.09
	60	48.33	52.73	56.67	61.82	58.33	63.64	66.67	72.73	68.33	74.55	66.67	72.73	58.33	63.64
	70	51.43	65.45	54.29	69.09	55.71	70.91	64.29	81.82	60.00	76.36	62.86	80.00	55.71	70.91
	91	46.15	76.36	48.35	80.00	48.35	80.00	51.65	85.45	52.75	87.27	60.44	100.00	47.25	78.18
	127	40.94	94.55	41.73	96.36	39.37	90.91	43.31	100.00	41.73	96.36	43.31	100.00	41.73	96.36
	145	37.93	100.00	37.24	98.18	37.24	98.18	37.93	100.00	37.24	98.18	37.93	100.00	36.55	96.36
	162	33.95	100.00	33.95	100.00	33.33	98.18	33.95	100.00	33.33	98.18	33.95	100.00	33.33	98.18
	164	33.54	100.00	33.54	100.00	32.93	98.18	33.54	100.00	32.93	98.18	33.54	100.00	33.54	100.00
	186	29.57	100.00	29.57	100.00	29.57	100.00	29.57	100.00	29.03	98.18	29.57	100.00	29.57	100.00
	187	29.41	100.00	29.41	100.00	29.41	100.00	29.41	100.00	29.41	100.00	29.41	100.00	29.41	100.00
	Average	41.71	77.02	43.39	78.51	45.61	79.50	48.44	83.31	48.16	82.31	49.11	84.46	46.18	80.00
Sonar	5	80.00	40.00	80.00	40.00	100.00	50.00	80.00	40.00	100.00	50.00	100.00	50.00	100.00	50.00
	7	85.71	60.00	85.71	60.00	85.71	60.00	71.43	50.00	85.71	60.00	85.71	60.00	100.00	70.00
	9	77.78	70.00	77.78	70.00	77.78	70.00	77.78	70.00	77.78	70.00	88.89	80.00	88.89	80.00
	11	72.73	80.00	72.73	80.00	81.82	90.00	81.82	90.00	81.82	90.00	72.73	80.00	90.91	100.00
	12	66.67	80.00	75.00	90.00	83.33	100.00	83.33	100.00	83.33	100.00	75.00	90.00	83.33	100.00
	13	69.23	90.00	69.23	90.00	76.92	100.00	76.92	100.00	76.92	100.00	76.92	100.00	76.92	100.00
	14	64.29	90.00	71.43	100.00	71.43	100.00	71.43	100.00	71.43	100.00	71.43	100.00	71.43	100.00
	15	66.67	100.00	66.67	100.00	66.67	100.00	66.67	100.00	66.67	100.00	66.67	100.00	66.67	100.00
	Average	72.88	76.25	74.82	78.75	80.46	83.75	76.17	81.25	80.46	83.75	79.67	82.50	84.77	87.50
Wdbc	10	90.00	23.08	90.00	23.08	100.00	25.64	100.00	25.64	90.00	23.08	100.00	25.64	100.00	25.64
	20	90.00	46.15	90.00	46.15	95.00	48.72	95.00	48.72	95.00	48.72	95.00	48.72	100.00	51.28
	30	93.33	71.79	93.33	71.79	96.67	74.36	96.67	74.36	96.67	74.36	96.67	74.36	96.67	74.36
	43	88.37	97.44	90.70	100.00	88.37	97.44	88.37	97.44	90.70	100.00	86.05	94.87	90.70	100.00
	44	86.36	97.44	88.64	100.00	86.36	97.44	88.64	100.00	88.64	100.00	86.36	97.44	88.64	100.00
	46	84.78	100.00	84.78	100.00	82.61	97.44	84.78	100.00	84.78	100.00	82.61	97.44	84.78	100.00
	50	78.00	100.00	78.00	100.00	76.00	97.44	78.00	100.00	78.00	100.00	78.00	100.00	78.00	100.00
	51	76.47	100.00	76.47	100.00	76.47	100.00	76.47	100.00	76.47	100.00	76.47	100.00	76.47	100.00
	Average	85.92	79.49	86.49	80.13	87.69	79.81	88.49	80.77	87.53	80.77	87.64	79.81	89.41	81.41
Wine	6	16.67	10.00	50.00	30.00	33.33	20.00	16.67	10.00	16.67	10.00	0.00	0.00	66.67	40.00
	10	20.00	20.00	40.00	40.00	30.00	30.00	20.00	20.00	20.00	20.00	20.00	20.00	50.00	50.00
	20	30.00	60.00	40.00	80.00	40.00	80.00	30.00	60.00	30.00	60.00	30.00	60.00	40.00	80.00
	23	26.09	60.00	43.48	90.00	34.78	80.00	30.43	70.00	30.43	70.00	30.43	70.00	39.13	90.00
	24	25.00	60.00	41.67	100.00	33.33	80.00	29.17	70.00	29.17	70.00	29.17	70.00	41.67	100.00
	29	24.14	70.00	34.48	100.00	34.48	100.00	24.14	70.00	24.14	70.00	24.14	70.00	34.48	100.00
	47	17.02	80.00	21.28	100.00	21.28	100.00	21.28	100.00	17.02	80.00	17.02	80.00	21.28	100.00
	49	16.33	80.00	20.41	100.00	20.41	100.00	20.41	100.00	16.33	80.00	20.41	100.00	20.41	100.00
	54	14.81	80.00	18.52	100.00	18.52	100.00	18.52	100.00	18.52	100.00	18.52	100.00	18.52	100.00
	88	11.36	100.00	11.36	100.00	11.36	100.00	11.36	100.00	11.36	100.00	11.36	100.00	11.36	100.00
	Average	20.14	62.00	31.68	84.00	27.75	79.00	22.20	70.00	21.36	66.00	20.11	67.00	34.35	86.00

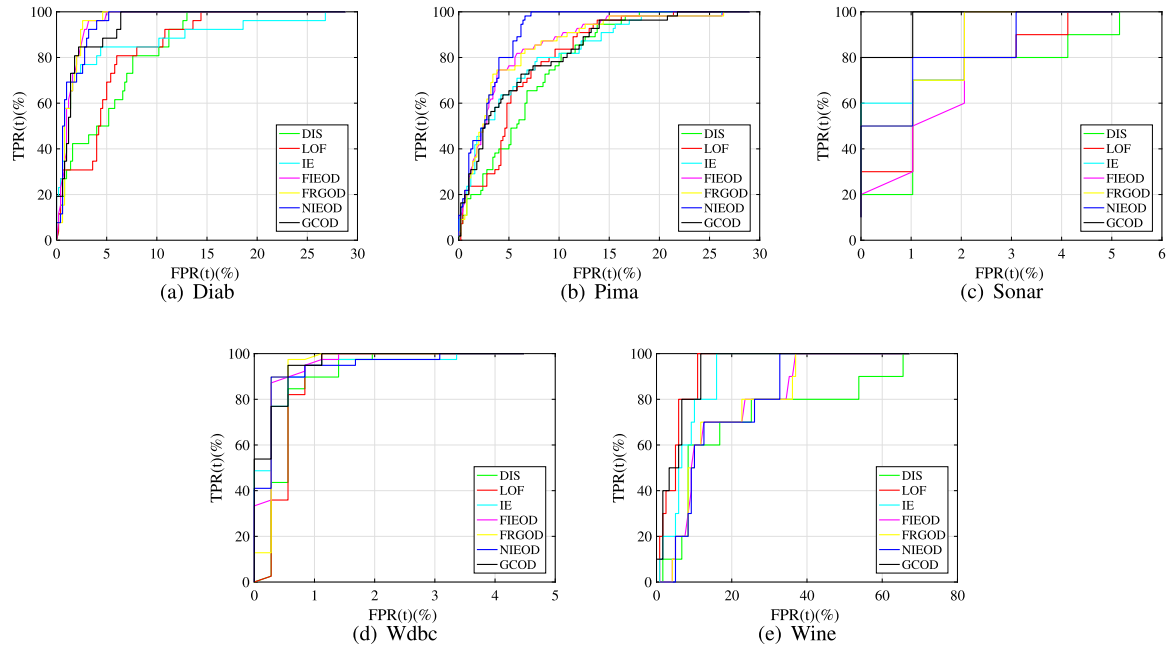


Fig. 1. ROC curves of numerical attribute datasets.

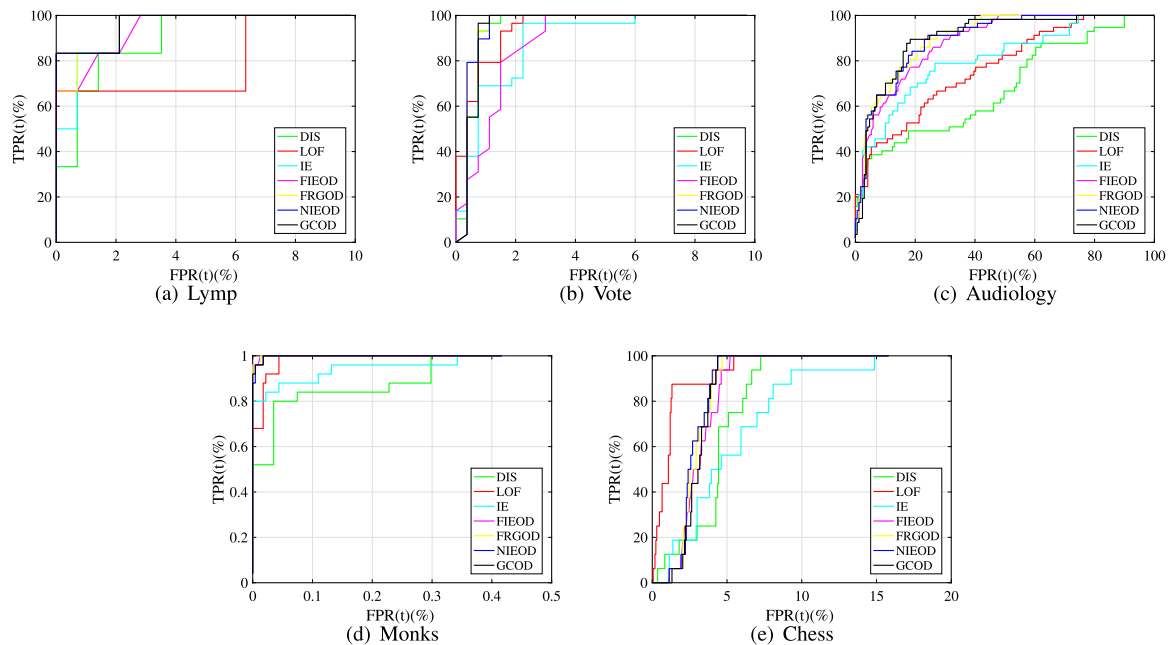


Fig. 2. ROC curves of categorical attribute datasets.

highest AUC values in only 1, 1, 5 and 3 datasets. On the other hand, the average AUC value of algorithm GCOD is 97.86%, which is greater than the other comparison algorithms. This shows that our proposed algorithm GCOD is effective and advantageous.

In summary, the experimental results show that our proposed method is superior in terms of performance and running time compared to some classical and latest methods based on the most of the experimental datasets. Specifically, algorithms FIEOD

and NIEOD are second only to our proposed algorithm GCOD. Through Table 11, we can conclude that algorithm GCOD has a much lower running time than algorithms FIEOD and NIEOD. In addition, algorithm GCOD does not require parameters, while algorithms LOF, FIEOD, FRGOD and NIEOD require parameters for numerical and mixed attribute datasets. These algorithms have to perform optimal parameter selection prior to the experiment, which can be time consuming.

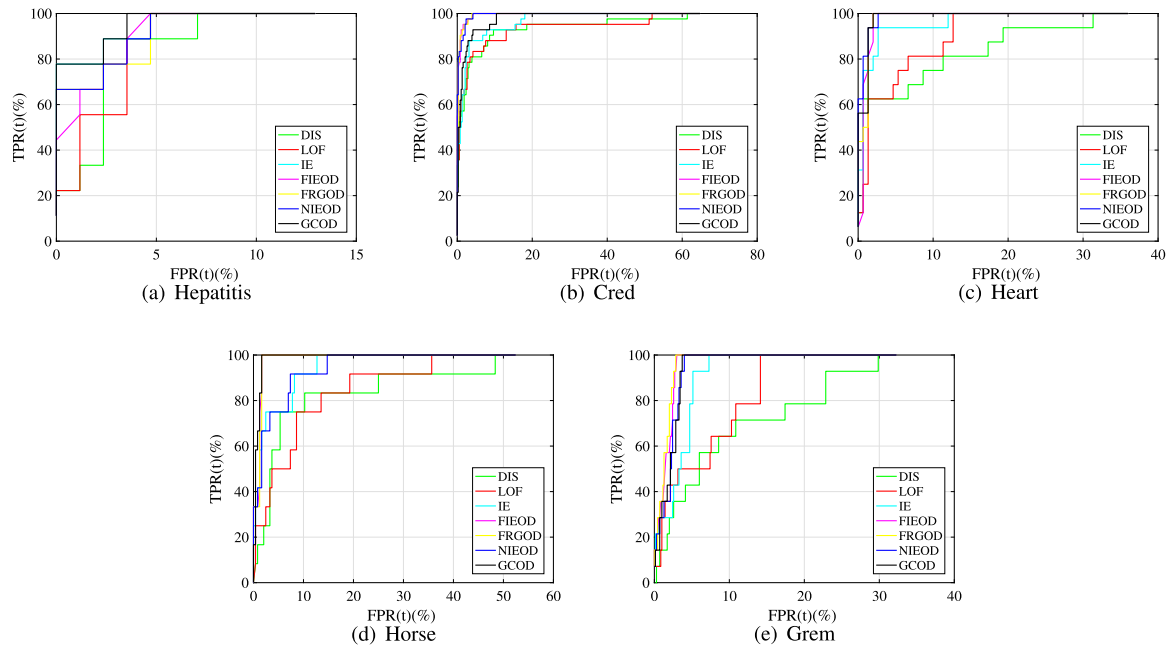


Fig. 3. ROC curves of mixed attribute datasets.

Table 13

Comparison of experimental results for categorical attribute datasets (%).

Data Set	t	DIS		LOF		IE		FIEOD		FRGOD		NIEOD		GCOD	
		P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)
Lymph	1	100.00	16.67	100.00	16.67	100.00	16.67	100.00	16.67	100.00	16.67	100.00	16.67	100.00	16.67
	3	66.67	33.33	100.00	50.00	100.00	50.00	100.00	50.00	100.00	50.00	100.00	50.00	100.00	50.00
	5	80.00	66.67	80.00	66.67	80.00	66.67	80.00	66.67	80.00	66.67	100.00	83.33	100.00	83.33
	7	71.43	83.33	57.14	66.67	71.43	83.33	71.43	83.33	71.43	83.33	71.43	83.33	71.43	83.33
	9	55.56	83.33	44.44	66.67	66.67	100.00	55.56	83.33	66.67	100.00	66.67	100.00	66.67	100.00
	11	54.55	100.00	36.36	66.67	54.55	100.00	54.55	100.00	54.55	100.00	54.55	100.00	54.55	100.00
	15	40.00	100.00	40.00	100.00	40.00	100.00	40.00	100.00	40.00	100.00	40.00	100.00	40.00	100.00
	Average	66.89	69.05	65.42	61.90	73.23	73.81	71.65	71.43	73.23	73.81	76.09	76.19	76.09	76.19
Vote	10	90.00	31.03	100.00	34.48	90.00	31.03	80.00	27.59	90.00	31.03	90.00	31.03	90.00	31.03
	20	90.00	62.07	90.00	62.07	90.00	62.07	80.00	55.17	95.00	65.52	95.00	65.52	90.00	62.07
	30	90.00	93.10	86.67	89.66	80.00	82.76	73.33	75.86	90.00	93.10	90.00	93.10	93.33	96.55
	32	87.50	96.55	84.38	93.10	81.25	89.66	75.00	82.76	93.10	93.33	93.10	93.33	90.63	100.00
	33	87.88	100.00	84.85	96.55	81.82	93.10	75.76	86.21	87.88	100.00	87.88	100.00	87.88	100.00
	35	82.86	100.00	82.86	100.00	80.00	96.55	77.14	93.10	82.86	100.00	82.86	100.00	82.86	100.00
	45	64.44	100.00	64.44	100.00	64.44	100.00	62.22	96.55	64.44	100.00	64.44	100.00	64.44	100.00
	50	58.00	100.00	58.00	100.00	58.00	100.00	58.00	100.00	58.00	100.00	58.00	100.00	58.00	100.00
	Average	81.34	85.34	81.40	84.48	78.19	81.90	72.68	77.16	82.27	85.34	82.27	85.34	82.14	86.21
Audiology	50	50.00	43.86	54.00	47.37	62.00	54.39	64.00	56.14	72.00	63.16	74.00	64.91	74.00	64.91
	120	30.00	63.16	37.50	78.95	39.17	82.46	39.17	82.46	45.00	94.74	45.00	94.74	46.67	98.25
	128	29.69	66.67	35.94	80.70	36.72	82.46	36.72	82.46	44.53	100.00	42.97	96.49	43.75	98.25
	151	32.45	85.96	33.77	89.47	33.11	87.72	33.11	87.72	37.75	100.00	37.75	100.00	37.09	98.25
	182	28.02	89.47	30.77	98.25	30.77	98.25	28.57	91.23	31.32	100.00	31.32	100.00	31.32	100.00
	183	28.42	91.23	30.60	98.25	31.15	100.00	28.96	92.98	31.15	100.00	31.15	100.00	31.15	100.00
	186	28.49	92.98	30.65	100.00	30.65	100.00	30.11	98.25	30.65	100.00	30.65	100.00	30.65	100.00
	187	28.34	92.98	30.48	100.00	30.48	100.00	30.48	100.00	30.48	100.00	30.48	100.00	30.48	100.00
	209	27.27	100.00	27.27	100.00	27.27	100.00	27.27	100.00	27.27	100.00	27.27	100.00	27.27	100.00
	Average	31.41	80.70	34.55	88.11	35.70	89.47	35.38	87.91	38.90	95.32	38.95	95.13	39.15	95.52
Monks	5	100.00	20.00	100.00	20.00	100.00	20.00	100.00	20.00	100.00	20.00	100.00	20.00	100.00	20.00
	10	100.00	40.00	100.00	40.00	100.00	40.00	100.00	40.00	100.00	40.00	100.00	40.00	100.00	40.00
	25	68.00	68.00	84.00	84.00	80.00	80.00	100.00	100.00	96.00	96.00	96.00	96.00	96.00	96.00
	28	71.43	80.00	82.14	92.00	75.00	84.00	89.29	100.00	89.29	100.00	85.71	96.00	85.71	96.00
	29	68.97	80.00	79.31	92.00	72.41	84.00	86.21	100.00	86.21	100.00	86.21	100.00	86.21	100.00
	35	57.14	80.00	71.43	100.00	62.86	88.00	71.43	100.00	71.43	100.00	71.43	100.00	71.43	100.00
	93	26.88	100.00	26.88	100.00	25.81	96.00	26.88	100.00	26.88	100.00	26.88	100.00	26.88	100.00
	103	24.27	100.00	24.27	100.00	24.27	100.00	24.27	100.00	24.27	100.00	24.27	100.00	24.27	100.00
	Average	64.59	71.00	71.00	78.50	67.54	74.00	74.76	82.50	74.26	82.00	73.81	81.50	73.81	81.50

(continued on next page)

Table 13 (continued).

Data Set	t	DIS		LOF		IE		FIEOD		FRGOD		NIEOD		GCOD	
		P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)
Chess	30	6.67	12.50	33.33	62.50	10.00	18.75	10.00	18.75	3.33	6.25	3.33	6.25	3.33	6.25
	89	12.36	68.75	16.85	93.75	10.11	56.25	7.87	43.75	16.85	93.75	16.85	93.75	17.98	100.00
	94	11.70	68.75	15.96	93.75	9.57	56.25	7.45	43.75	17.02	100.00	17.02	100.00	17.02	100.00
	107	11.21	75.00	14.95	100.00	8.41	56.25	6.54	43.75	14.95	100.00	14.95	100.00	14.95	100.00
	137	11.68	100.00	11.68	100.00	8.76	75.00	8.76	75.00	11.68	100.00	11.68	100.00	11.68	100.00
	254	6.30	100.00	6.30	100.00	5.91	93.75	6.30	100.00	6.30	100.00	6.30	100.00	6.30	100.00
	264	6.06	100.00	6.06	100.00	6.06	100.00	6.06	100.00	6.06	100.00	6.06	100.00	6.06	100.00
	Average	9.43	75.00	15.02	92.86	8.40	65.18	7.57	60.71	10.89	85.71	10.89	85.71	11.05	86.61

Table 14

Comparison of experimental results for mixed attribute datasets (%).

Data Set	t	DIS		LOF		IE		FIEOD		FRGOD		NIEOD		GCOD	
		P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)	P(t)	R(t)
Hepatitis	1	100.00	11.11	100.00	11.11	100.00	11.11	100.00	11.11	100.00	11.11	100.00	11.11	100.00	11.11
	3	66.67	22.22	66.67	22.22	100.00	33.33	100.00	33.33	100.00	33.33	100.00	33.33	100.00	33.33
	5	60.00	33.33	80.00	44.44	100.00	55.56	80.00	44.44	100.00	55.56	100.00	55.56	100.00	55.56
	7	71.43	55.56	71.43	55.56	100.00	77.78	71.43	55.56	85.71	66.67	85.71	66.67	100.00	77.78
	9	77.78	77.78	66.67	66.67	77.78	77.78	77.78	77.78	77.78	77.78	77.78	77.78	77.78	77.78
	12	66.67	88.89	66.67	88.89	66.67	88.89	75.00	100.00	66.67	88.89	66.67	88.89	75.00	100.00
	13	61.54	88.89	69.23	100.00	69.23	100.00	69.23	100.00	69.23	100.00	69.23	100.00	69.23	100.00
	15	60.00	100.00	60.00	100.00	60.00	100.00	60.00	100.00	60.00	100.00	60.00	100.00	60.00	100.00
	Average	70.51	59.72	72.58	61.11	84.21	68.06	79.18	65.28	82.42	66.67	82.42	66.67	85.25	69.44
Cred	10	100.00	23.81	100.00	23.81	100.00	23.81	100.00	23.81	100.00	23.81	100.00	23.81	90.00	21.43
	30	80.00	57.14	86.67	61.90	83.33	59.52	96.67	69.05	96.67	69.05	96.67	69.05	86.67	61.90
	57	59.65	80.95	61.40	83.33	64.91	88.10	71.93	97.62	73.68	100.00	71.93	97.62	68.42	92.86
	58	58.62	80.95	60.34	83.33	63.79	88.10	70.69	97.62	72.41	100.00	72.41	100.00	67.24	92.86
	72	52.78	90.48	51.39	88.10	54.17	92.86	58.33	100.00	58.33	100.00	58.33	100.00	54.17	92.86
	82	47.56	92.86	45.12	88.10	47.56	92.86	51.22	100.00	51.22	100.00	51.22	100.00	51.22	100.00
	111	36.04	95.24	36.04	95.24	37.84	100.00	37.84	100.00	37.84	100.00	37.84	100.00	37.84	100.00
	241	17.01	97.62	17.43	100.00	17.43	100.00	17.43	100.00	17.43	100.00	17.43	100.00	17.43	100.00
	277	15.16	100.00	15.16	100.00	15.16	100.00	15.16	100.00	15.16	100.00	15.16	100.00	15.16	100.00
	Average	51.87	79.89	52.62	80.42	53.80	82.80	57.70	87.57	58.08	88.10	57.89	87.83	54.24	84.66
Heart	5	80.00	25.00	80.00	25.00	100.00	31.25	80.00	25.00	100.00	31.25	100.00	31.25	100.00	31.25
	10	90.00	56.25	80.00	50.00	90.00	56.25	90.00	56.25	80.00	50.00	100.00	62.50	90.00	56.25
	19	52.63	62.50	57.89	68.75	78.95	93.75	78.95	93.75	78.95	93.75	78.95	93.75	84.21	100.00
	20	50.00	62.50	60.00	75.00	75.00	93.75	80.00	100.00	80.00	100.00	80.00	100.00	80.00	100.00
	34	38.24	81.25	44.12	93.75	47.06	100.00	47.06	100.00	47.06	100.00	47.06	100.00	47.06	100.00
	35	37.14	81.25	45.71	100.00	45.71	100.00	45.71	100.00	45.71	100.00	45.71	100.00	45.71	100.00
	63	25.40	100.00	25.40	100.00	25.40	100.00	25.40	100.00	25.40	100.00	25.40	100.00	25.40	100.00
	Average	53.34	66.96	56.16	73.21	66.02	82.14	63.87	82.14	65.30	82.14	68.16	83.93	67.48	83.93
Horse	5	40.00	16.67	60.00	25.00	80.00	33.33	80.00	33.33	80.00	33.33	80.00	33.33	80.00	33.33
	10	30.00	25.00	40.00	33.33	70.00	58.33	70.00	58.33	70.00	58.33	60.00	50.00	80.00	66.67
	16	43.75	58.33	37.50	50.00	56.25	75.00	75.00	100.00	75.00	100.00	50.00	66.67	75.00	100.00
	43	23.26	83.33	23.26	83.33	27.91	100.00	27.91	100.00	27.91	100.00	25.58	91.67	27.91	100.00
	48	20.83	83.33	20.83	83.33	25.00	100.00	25.00	100.00	25.00	100.00	25.00	100.00	25.00	100.00
	99	11.11	91.67	12.12	100.00	12.12	100.00	12.12	100.00	12.12	100.00	12.12	100.00	12.12	100.00
	130	9.23	100.00	9.23	100.00	9.23	100.00	9.23	100.00	9.23	100.00	9.23	100.00	9.23	100.00
	Average	25.45	65.48	28.99	67.86	40.07	80.95	42.75	84.52	42.75	84.52	37.42	77.38	44.18	85.71
Grem	12	16.67	14.29	33.33	28.57	33.33	28.57	41.67	35.71	41.67	35.71	33.33	28.57	41.67	35.71
	24	20.83	35.71	25.00	42.86	25.00	42.86	29.17	50.00	41.67	71.43	33.33	57.14	33.33	57.14
	34	14.71	35.71	20.59	50.00	23.53	57.14	38.24	92.86	41.18	100.00	32.35	78.57	32.35	78.57
	36	16.67	42.86	19.44	50.00	22.22	57.14	38.89	100.00	38.89	100.00	33.33	85.71	33.33	85.71
	40	15.00	42.86	17.50	50.00	20.00	57.14	35.00	100.00	35.00	100.00	32.50	92.86	35.00	100.00
	42	14.29	42.86	16.67	50.00	21.43	64.29	33.33	100.00	33.33	100.00	33.33	100.00	33.33	100.00
	65	12.31	57.14	13.85	64.29	21.54	100.00	21.54	100.00	21.54	100.00	21.54	100.00	21.54	100.00
	113	8.85	71.43	12.39	100.00	12.39	100.00	12.39	100.00	12.39	100.00	12.39	100.00	12.39	100.00
	223	6.28	100.00	6.28	100.00	6.28	100.00	6.28	100.00	6.28	100.00	6.28	100.00	6.28	100.00
	Average	13.95	49.21	18.34	59.52	20.64	67.46	28.50	86.51	30.22	89.68	26.49	82.54	27.69	84.13

5. Conclusion

Outlier detection is becoming more and more important in lots of fields such as intrusion detection, public safety, and medical diagnosis, and so on. In this paper, we introduce formal concept analysis into outlier detection and propose a granular concept-based outlier detection model based on a distance measure between granular concepts. And then a related algo-

rithm is designed by this model. Finally, some datasets are chosen from publicly available web pages for comparison experiments with some existing classical and rough set-based outlier detection algorithms. The experimental results indicate that the proposed algorithm is effective and efficient under most of the datasets. In future research, we will study outlier detection methods based on other concept lattice models, such as object-oriented, attribute-oriented, three-way, fuzzy concept lattices, and so on.

Table 15
The AUC value (%).

	DIS	LOF	IE	FIEOD	FRGOD	NIEOD	GCOD
Diab	95.15	95.36	96.47	98.89	98.67	98.60	98.28
Pima	93.72	94.44	94.89	95.61	95.93	97.37	94.69
Sonar	98.35	98.66	99.28	98.66	99.18	99.07	99.79
Wdbc	99.45	99.48	99.69	99.71	99.72	99.70	99.78
Wine	79.66	95.04	92.27	85.63	84.79	84.87	94.87
Lymph	98.94	97.89	99.41	99.18	99.53	99.65	99.65
Vote	99.46	99.43	98.86	98.40	99.45	99.52	99.46
Audiology	68.17	77.06	80.87	80.22	90.25	89.79	90.12
Monks	94.23	99.23	97.40	100.00	99.95	99.89	99.91
Chess	95.75	98.72	94.76	94.01	97.12	97.21	96.98
Hepatitis	97.65	97.91	99.22	99.22	98.69	98.82	99.22
Cred	95.43	95.45	97.61	99.56	99.61	99.60	98.49
Heart	93.75	96.08	98.50	99.08	99.25	99.35	99.38
Horse	90.78	91.36	96.96	99.15	99.04	96.82	99.28
Germ	90.30	93.70	96.83	98.30	98.66	98.01	97.99
Average	92.72	95.32	96.20	96.37	97.32	97.22	97.86

CRedit authorship contribution statement

Qian Hu: Conceptualization, Methodology, Data curation, Writing – original draft. **Zhong Yuan:** Data curation, Writing – review & editing. **Keyun Qin:** Writing – review & editing, Funding acquisition. **Jun Zhang:** Writing – review & editing, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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